

Authoritative Data Repository Capability Proposal

A sanitized example of enterprise GIS capability planning

BUSINESS PROBLEM

Geospatial data is fragmented, difficult to locate, and inconsistently managed. Project teams spend time rebuilding information, customers cannot rely on a common source, and decision-makers lack a dependable view of installation conditions.

PROPOSED CAPABILITY

Create a living Authoritative Data Repository that centralizes installation GIS data, controls access, tracks data quality, and supports internal delivery as well as customer-facing data services.

EXPECTED VALUE

Reliable data reduces duplicated effort and improves project decisions. The same capability creates a foundation for managed data services, spatial analysis, facility planning, and future investment prioritization.

Operating Scope

The repository connects data management, technology, work processes, and authorized users

BUSINESS AND PROCESS

Collect, validate, manage, store, and deliver installation GIS data for project teams and customers. Incorporate updates from field collection, project delivery, and contractor submissions.

DATA

Core holdings include cadastral, planning, facility, infrastructure, utility, environmental, survey, as-built, elevation, and real property information.

TECHNOLOGY

Use an enterprise geodatabase within an approved GIS portal. A central application provides controlled access to data, maps, applications, plans, drawings, and supporting records.

PEOPLE AND ACCESS

Role-based permissions separate viewing, editing, stewardship, and administration. Project teams gain a trusted source while customers receive managed information and analysis suited to their decisions.

GIS Capability Development

The repository supports increasingly advanced decisions as coverage and quality improve

DESCRIBE

Map assets, facilities, infrastructure, utilities, and environmental conditions. Compare status, condition, and constraints across locations.

DIAGNOSE

Identify where facility actions are needed. Evaluate project locations, utility routes, environmental impacts, and other operational constraints.

PREDICT

Forecast facility, utility, and project needs. Estimate where future demand or environmental effects may occur.

PRESCRIBE

Recommend project locations, facility actions, utility upgrades, and mitigation measures using documented criteria and current data.

Capability maturity depends on trustworthy data, clearly assigned stewardship, and continuous feedback from the people who use the information.

Implementation and Evidence of Success

A phased approach ties repository growth to demand, governance, and measurable results

IMPLEMENTATION

1. Consolidate available planning datasets and document provenance.
2. Assess coverage, accuracy, completeness, and priority gaps.
3. Establish stewardship, permissions, update procedures, and acceptance criteria.
4. Expand through field collection and project delivery.
5. Add advanced analysis when data and business demand support it.

SUCCESS MEASURES

- Users can locate authoritative data without searching disconnected drives.
- Project teams reuse current information instead of rebuilding datasets.
- Quality issues have assigned owners and documented resolution.
- Customers receive dependable maps, data services, and analyses.
- Leadership can connect repository investment to project quality, efficiency, and service demand.

Key assumptions: confirmed demand, engaged stakeholders, sustained leadership support, adequate funding, and access limited to authorized users.